

Sai Prasad E

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PROFESSIONAL SUMMARY

Senior Data Engineer with 3+ years designing and operating large-scale lakehouse, identity resolution, and Customer 360 platforms on Azure Databricks for a global retail client — processing 5M+ PII records daily across a 197M-customer base. Led a 6-member team through a vendor re-architecture that cut match latency by 66% and vendor API volume by 95%. Currently building an LLM-orchestrated data quality agent on Google Vertex AI for cross-platform migration validation. Databricks-certified (Data Engineer Associate, Spark Developer Associate) and Azure DP-203 certified.

TECHNICAL SKILLS

Languages: Python, SQL (Spark SQL, Snowflake SQL, T-SQL)

Big Data & Lakehouse: PySpark, Delta Lake, Structured Streaming, Change Data Feed (CDF), MERGE/SCD2 patterns, Unity Catalog, Liquid Clustering, dbt (models, macros, tests), Databricks Genie

Cloud Platforms: Azure Databricks, Azure Data Factory, Azure Synapse, Azure Key Vault, GCP BigQuery, Google Vertex AI

GenAI & Data Quality: LLM agent orchestration, governed SQL generation (sqlglot), automated reconciliation and RCA frameworks, data quality validation at scale

DevOps & Orchestration: Databricks Jobs API and Workflows, Jenkins, GitHub Actions, CI/CD, multi-environment deployments, Lightup, PagerDuty

Collaboration: Power BI (downstream), Jira, Confluence, client stakeholder management

PROFESSIONAL EXPERIENCE

TheMathCompany (MathCo), Bangalore

Senior Data Engineer

Jan 2023 – Present

July 2026 – Present

- Designing and building (solo) an LLM-orchestrated data quality agent on Google Vertex AI that automates validation of a Databricks-to-BigQuery migration across 6 pipeline stages: schema profiling, column inference, SQL rule generation, execution, root-cause analysis, and reporting.
- Architected a governed SQL generation layer where the LLM emits structured rule specifications — not raw SQL — compiled and safety-checked by a SQLCompiler and QueryGuard framework enforcing read-only execution and multi-dialect compatibility via sqlglot.
- Implemented confidence-gated approval workflows that auto-resolve column mappings and measure definitions above empirical thresholds while always routing primary keys to human review, preventing silent data contract violations.
- Built a YAML-based context layer generated from pipeline code and table metadata, enabling automated root-cause analysis (e.g., stale-load detection explaining count mismatches) and handling cross-platform data model changes such as SCD2 conversion and column additions/removals.

Cloud Engineer II

July 2025 – June 2026

- Led a 6-member team through a 6-month re-architecture of the identity resolution matching layer, replacing a legacy VM/file-based matching vendor with an API-based entity resolution service — owning requirement breakdown, task allocation, and design adaptation while delivering an individual workstream in parallel.
- Built the API integration layer from the ground up (request payload construction, response parsing and flattening, customer-to-vendor ID mapping), cutting end-to-end match latency from 3 hours to 1 hour and unblocking the client's near-real-time streaming roadmap.
- Designed an internal API-result caching layer serving previously matched records from an internal store, reducing daily vendor API calls from ~10M to ~500K (95% reduction) and directly cutting per-call vendor spend.
- Executed a one-time 197M-customer historical match consolidating ~20M duplicate profiles (~10% of the customer base); expanded deterministic matching with loyalty-ID and e-commerce-ID match keys, reducing daily duplicate profile creation by ~7% overall.
- Delivered a zero-disruption production cutover via a 1-week parallel run of legacy and new systems, validating match-rate parity and proactively communicating downstream impacts — earning direct client appreciation.

- Solo-delivered a 25-table migration of assortment localization data from a legacy notebook-based platform to the enterprise data platform in 1 month — redesigning notebook logic into modular dbt models on Databricks, live in production; engineered 9 reusable dbt macros parameterized by brand and market to generate all 25 models from shared logic.
- Built an automated reconciliation framework (leveraging Databricks Genie) reporting row counts, primary-key and uniqueness checks, null percentages, schema and column-mapping checks, and PK-level record diffs on tables up to 120M rows; root-caused every variance via tolerance-band bucketing and secured up-stream/downstream sign-off before cutover.
- Built 5 semantic-layer KPI tables (Loyalty Spend Summary, Total Sales Summary, CLTV) in Databricks bridging Customer 360 data and Power BI dashboards used by client executives to track offer-launch impact on sales and drive launch-timing decisions during peak periods.
- Owned client-facing requirement gathering through daily stakeholder calls — documenting requirements, driving gap-clarification cycles, and walking clients through delivered solutions for sign-off.

Cloud Engineer I

Oct 2023 – June 2025

- Engineered batch identity resolution pipelines on Azure Databricks (PySpark, Delta Lake) processing 5M+ PII records daily across 11 heterogeneous sources — creating 30–40K new customer profiles and enriching ~1M profiles per day for downstream marketing activation.
- Designed and built 3 attribute mappers (email, address) and owned end-to-end ingestion for 5 data sources, implementing source-specific precedence and business rules for profile creation across an SCD2 dimensional model (identity spine plus email, name, phone, and address dimensions).
- Enhanced deterministic match rules and integrated a third-party fuzzy-matching service for name and address resolution, improving match accuracy and reducing duplicate customer profiles versus the legacy system.
- Identified and remediated a production defect causing incorrect cross-brand profile creation — performed root-cause analysis, quantified impacted records, and executed the fix in production with zero downstream disruption.
- Extended the ingestion framework to accelerate new-source onboarding (e.g., SMS profiles) and built brand/market-level customer attributes and campaign-facing derived flags (e.g., emailability status).
- Built 5+ customer- and store-level Customer 360 unified views (loyalty, orders, transactions, aggregates, store distance) over datasets spanning 3–4B+ transaction records and a ~1B-row website clickstream view — consumed by Marketing, Analytics, BI (Power BI), and Data Science teams.
- Cut a critical view's refresh runtime from 30 minutes to 5 minutes (~83% reduction) by replacing full refresh with an incremental load strategy and enabling Delta liquid clustering.
- Implemented multi-brand data stitching and source-specific business-rule transformations (e.g., opt-in/opt-out derivation with per-source logic) across unified views.

Data Engineering Intern

Jan 2023 – May 2023

- Built a Market Mix Modeling framework and budget optimization tool in Python, automating data preparation, feature engineering, and scenario analysis for marketing analytics use cases.

EDUCATION

SASTRA University, Thanjavur

B.Tech in Electrical and Electronics Engineering

2019 – 2023

CGPA: 8.19/10

CERTIFICATIONS

- Databricks Certified Data Engineer Associate
- Databricks Certified Associate Developer for Apache Spark
- Microsoft Certified: Azure Data Engineer Associate (DP-203)
- Databricks Academy Accreditation: Platform Administrator